



DCA0305

morsinaldo.medeiros.075@ufrn.edu.br

ivanovitch.silva@ufrn.br

Machine Learning-Based Systems Design

Beginning
of the
Journey

Linear Algebra & Math

Matrices & vector arithmetics,
types, operations factorization,
derivatives

Data Science

EDA, measurements of
centrality (mean, mode,
median, variand, std, z-score),
data pipeline

Deep Learning

Fundamentals of Deep
Learning, Better Generalization
vs Better Learning,
Hyperparameter Tuning, Batch
Normalization, CNN, Transfer
Learning

Probability & Statistics

Cond. prob., distributions,
bayesian prob., data viz., central
limit theorem, hypothesis tests,
correlation, resampling
methods

Machine Learning

Supervised Learning (KNN, Linear
Regression, Logistic Regression, DT, RF,
Ensemble, XGBoost, MLP), Unsupervised
Learning (K-means, PCA)

TinyML/EdgeAI

Optimization,
quantization, deploy
into a microcontroller

But

If a model is not
deployed, it does not
generate value

New Models

Transformers, Difusion, GAN, LLM, GNN,
Generative Flow Networks, SLM/Domain
Specific (DSLIM), Agents, World Models,
Continual Learning, Physical AI ...

Cont.

2016

Pandas

2017

Scikit-Learn

2018-2020

TensorFlow, Keras, Pytorch,
Covid-19

2021-2022*

MLOps, GAN, LLM, GPT-3,
OpenAI

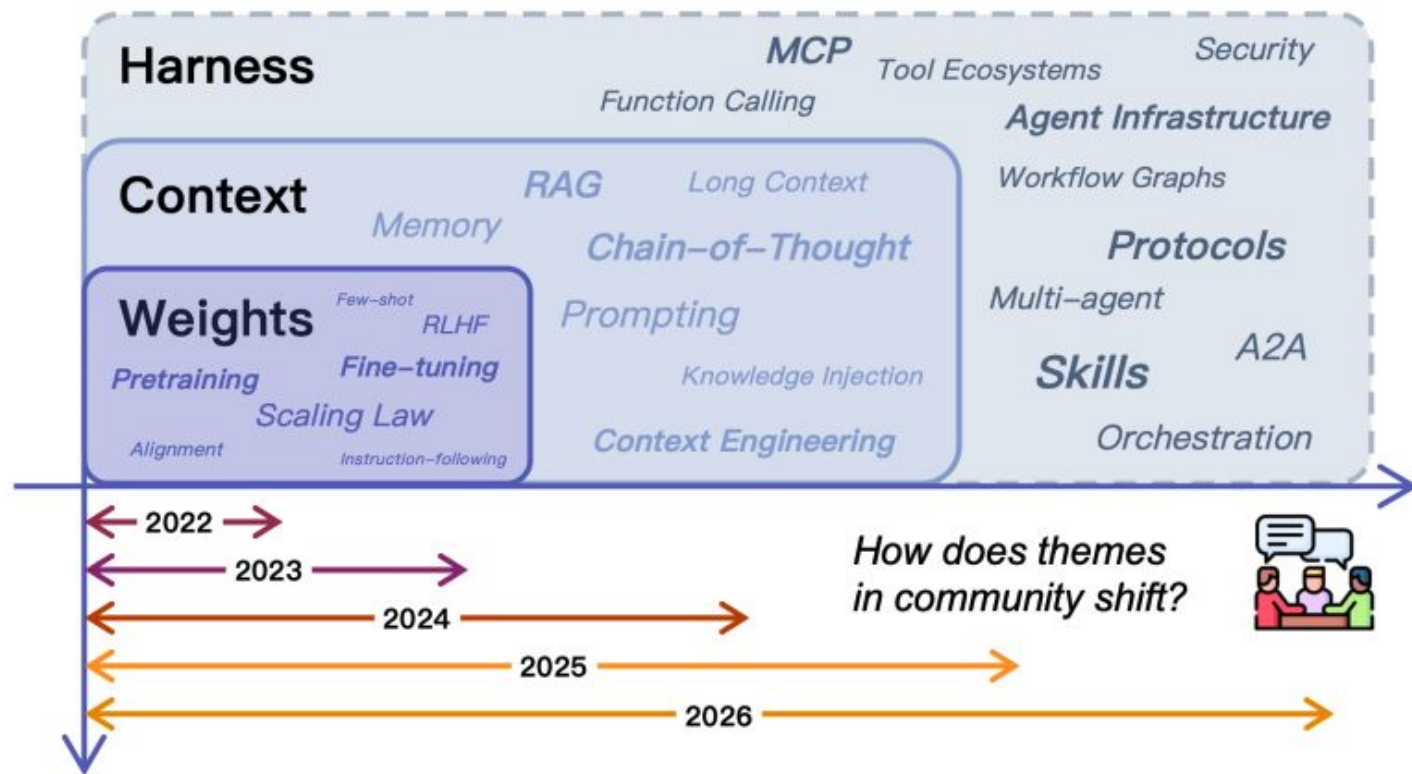
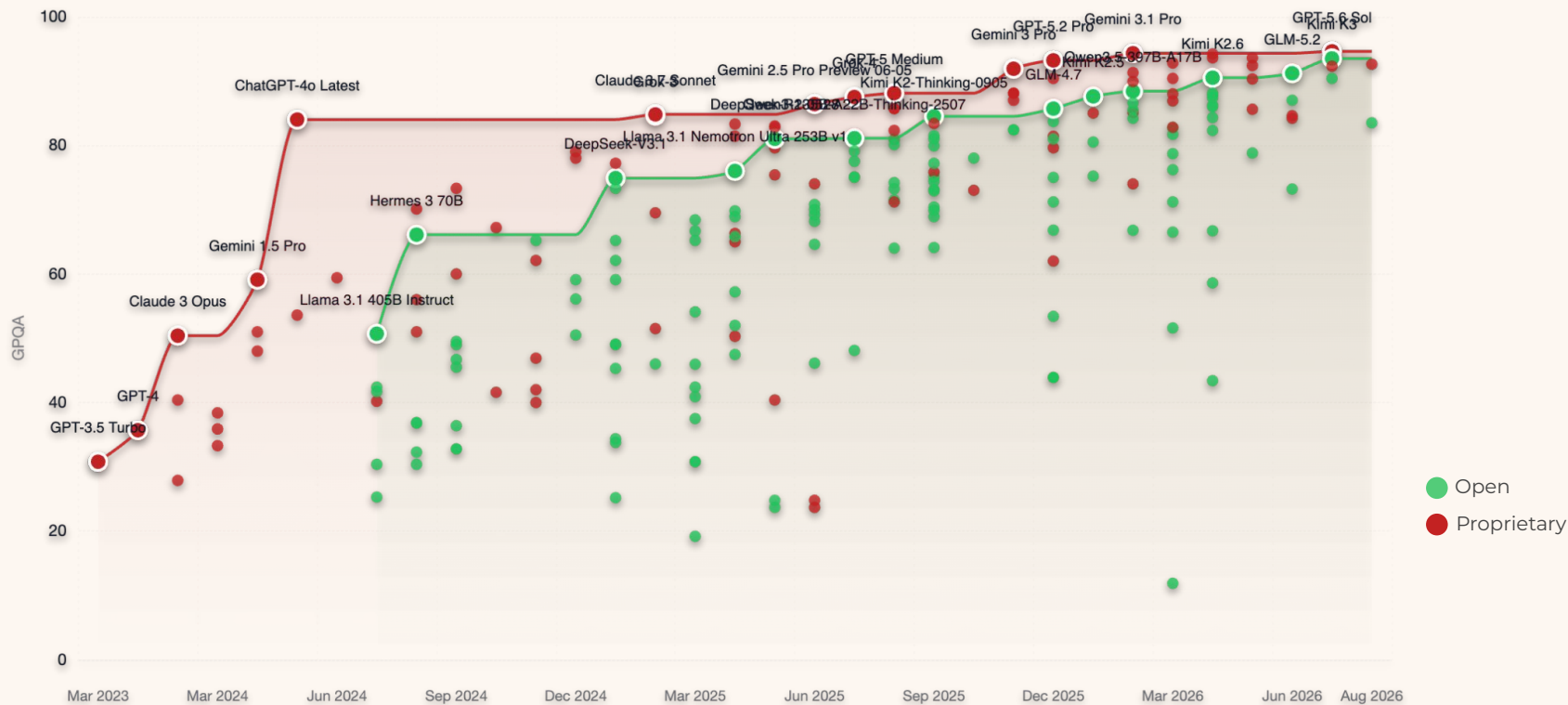
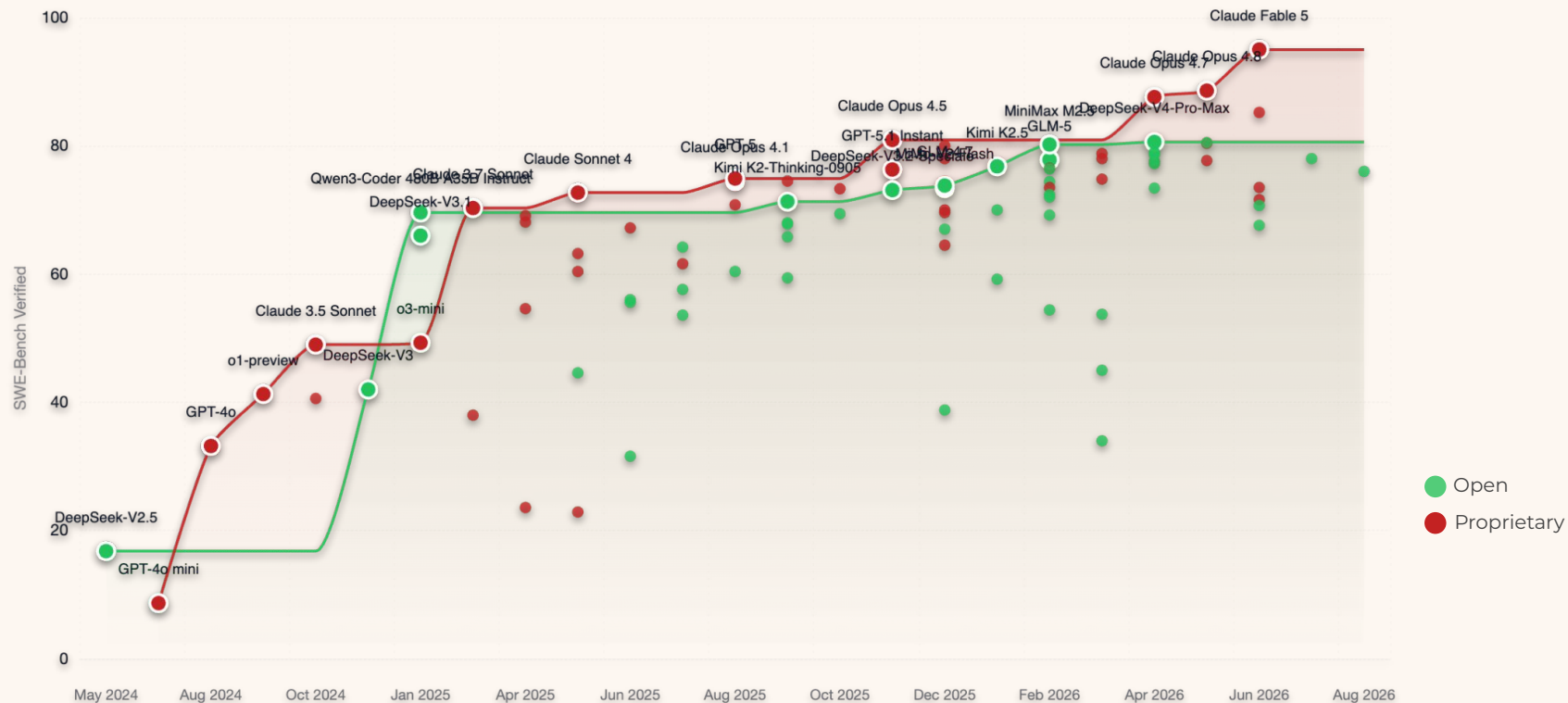


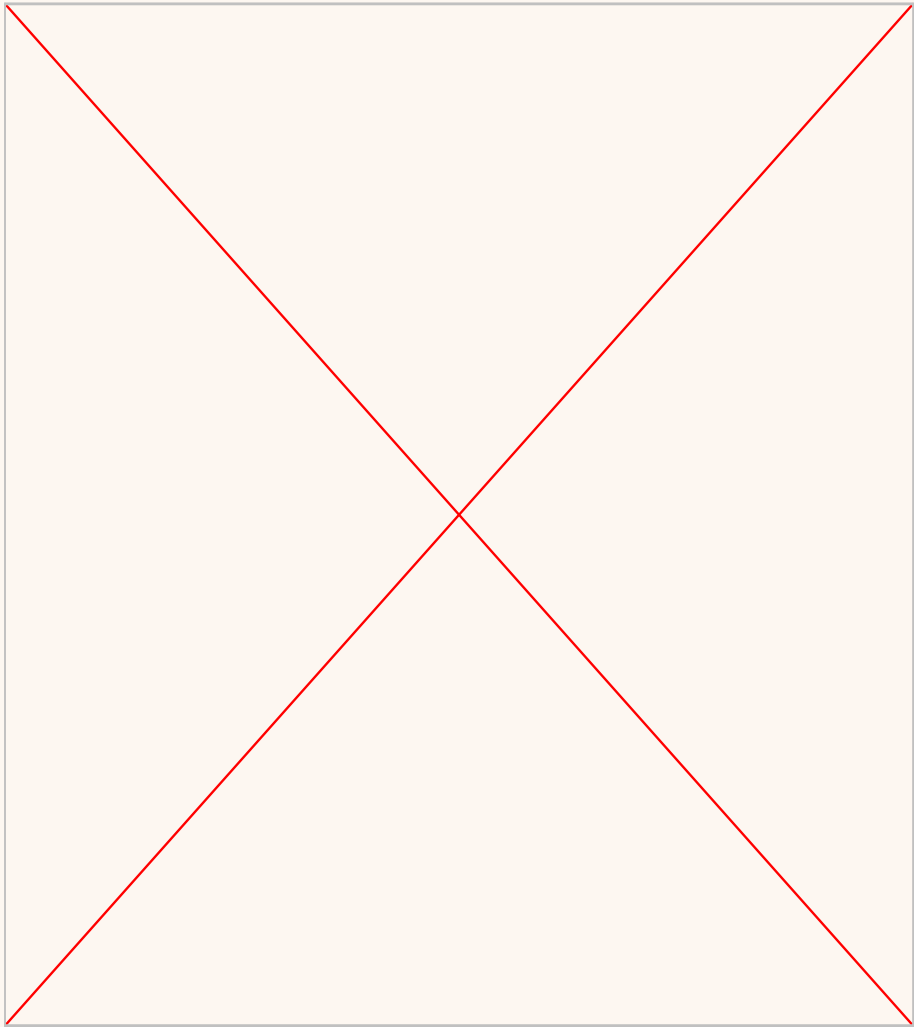
Figure 2 Community theme evolution across three capability layers. The stacked layers—Weights, Context, and Harness—show how the center of gravity in the LLM agent community has shifted outward over time, from parametric knowledge and prompting toward harness-level infrastructure such as tool ecosystems, protocols, skills, and multi-agent orchestration.

LLM Leaderboard & GPQA

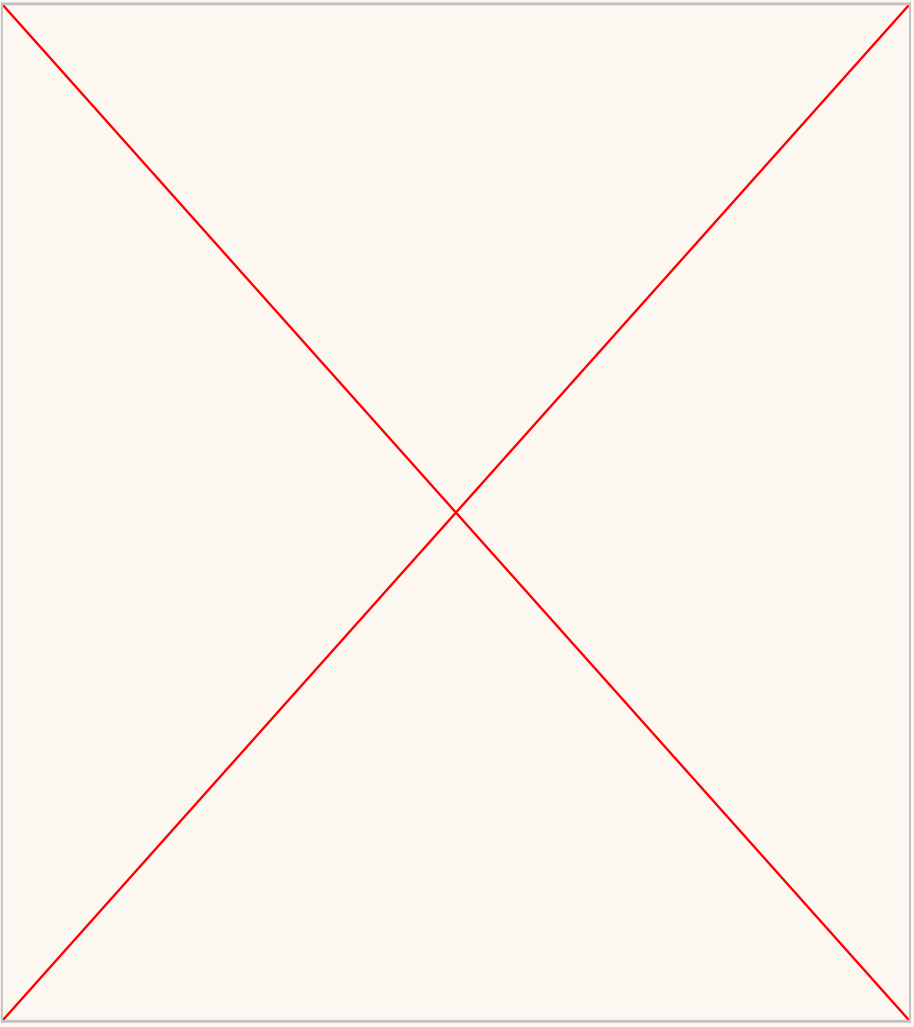


LLM Leaderboard & SWE-Bench





<https://www.instagram.com/p/DaoBWpJpCMO/>



<https://www.instagram.com/p/DaybjUsJppa/>

← Post



As engineering, product, design, DS, etc. melt into a new kind of role, I was reflecting on what roles might look like in the future. For example, when I look at the Claude Code team I see what I think is five archetypes:

1. **Prototyper**: comes up with brand new ideas; churns out many ideas, most of which don't ship
2. **Builder**: quickly turns a prototype/idea into production-grade product/infra
3. **Sweeper**: cleans up the UI, simplifies the code and system, unships, optimizes performance
4. **Grower**: takes a product that has been built and iterates on it to improve Product-Market Fit
5. **Maintainer**: owns a mature system to make it secure, reliable, fast, and efficient as it scales

Many people span across 2 roles, and sometimes 3 roles. I also notice that these roles are not really tied to job function -- eg. across Anthropic, some designers match category 1, some 2, some 3; same for engineers, PM, DS.

A healthy team needs a mix of these, depending on the product:

- A product that is new and pre-PMF needs people that are strong at 1+2+3
- A product that is growing and has found PMF needs 2+3+4 and some 5
- A product that has strong PMF needs 3+4+5 and some 2

Maybe product roles of the future will look more like this, and less like the domain-specific roles of today?

8:45 PM · Jun 28, 2026 · 3.1M Views



i never quite liked the idea of defining archetypes, because it's easy for people to look at it and think "ah that's who i am" and never question themselves again

in reality, a person's role often needs to evolve along with the project. when i'm starting a new project i'm a prototyper and builder, but very soon i become a sweeper when rough edges become the bottleneck. then i become a grower and maintainer as the project becomes more mature. if i box myself into a particular role, i would have to give up the project somewhere along the way

and increasingly, when we juggle between multiple projects at the same time, we need to be able to play different roles in different places. categorizing ourselves into certain archetypes often prevent people from expanding their ambition

so my advice - stay flexible and be obsessed with whatever is important for achieving your goal, and care less about the boundaries of roles that will continue to blur over time

8:59 PM · Jun 28, 2026 · 113.9K Views



Relevant ▾

View quotes >



Post your reply

Reply



Boris Cherny ✓ @bcherny · Jun 28

Totally agree. Roles often change over time/project



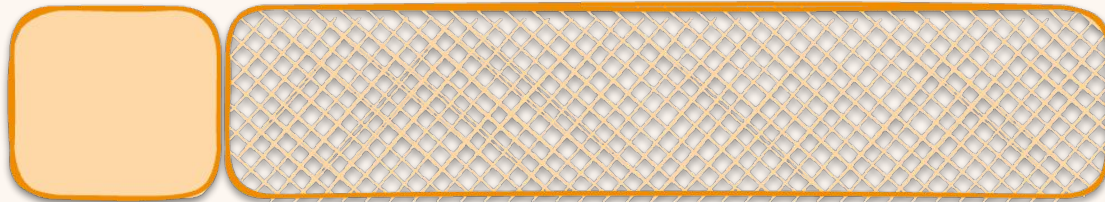
<https://x.com/bcherny/status/2071379474277613732>

What tasks should
you never do
manually again?

Thinking
& Hands-on

Writing &
Note-Taking

Before



Today



Paper/Project Submission

How to project a
typical machine
learning workflow?

O'REILLY®

May, 2022

Designing Machine Learning Systems

An Iterative Process
for Production-Ready
Applications



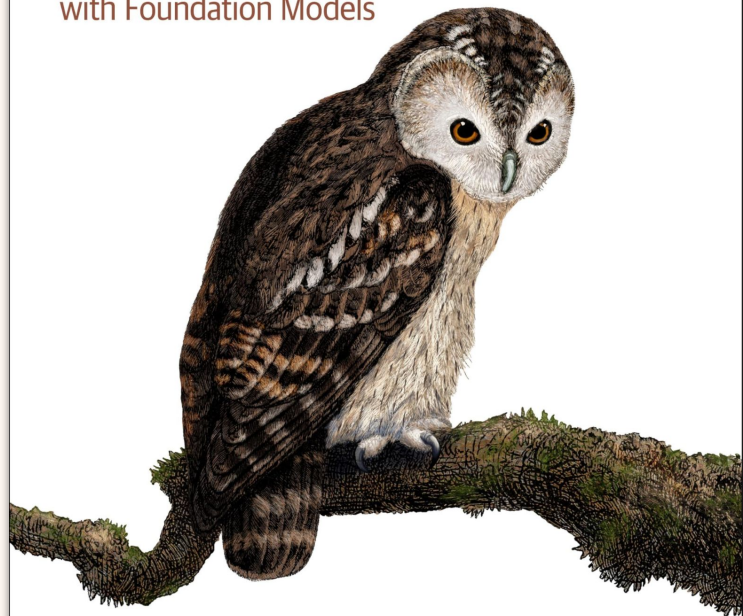
Chip Huyen

O'REILLY®

Jan., 2025

AI Engineering

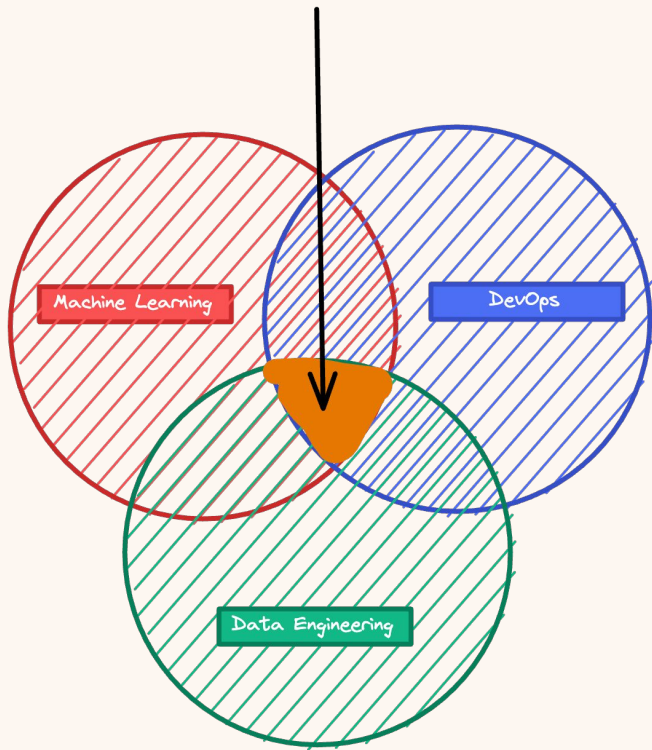
Building Applications
with Foundation Models



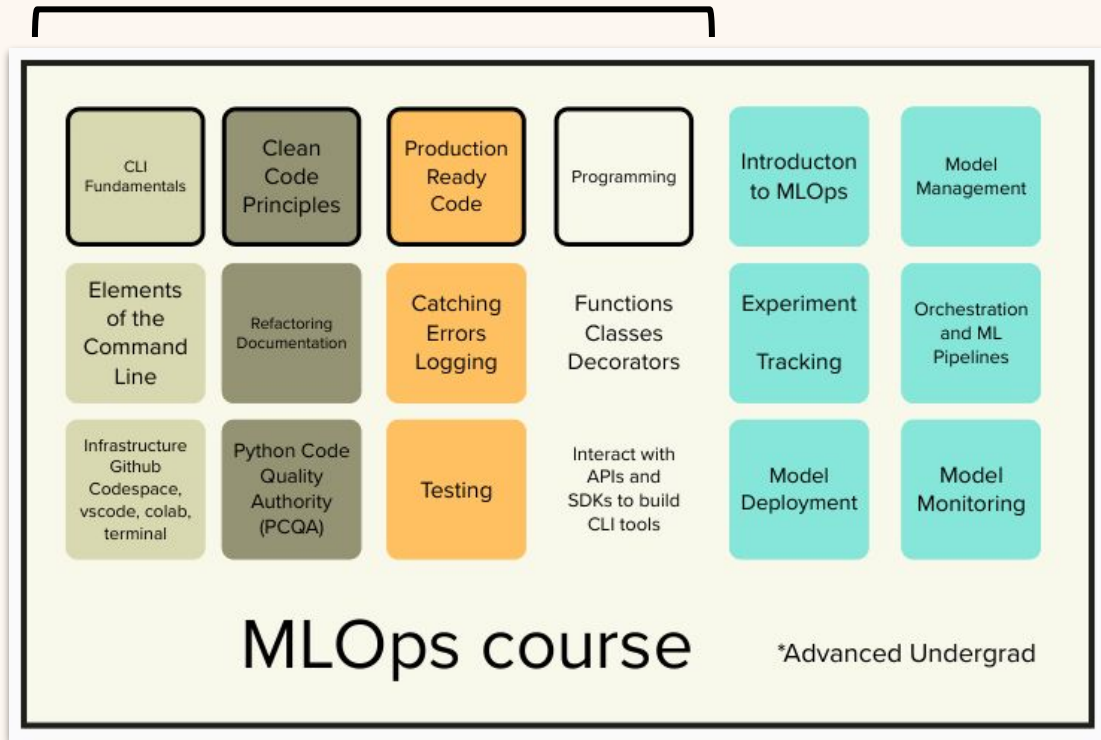
Chip Huyen

AI Engineering >> "The process of build applications on top of readily available models"

MLOps



Machine Learning/Deep Learning



1

Should I build this AI application? How do I evaluate any application? Can I use AI to evaluate AI output?

2

What causes hallucinations? How do I detect and mitigate hallucination?

3

What are the best practices for prompt engineering? Why does RAG work? What are the strategies for doing RAG?

4

What's an agent? How do I build and evaluate an agent?

5

When to finetune a model? When not to finetune a model? How much data do I need? How do I validate the quality of my data?

6

How do I make my model faster, cheaper, and secure? How do I create a feedback loop to improve my application continually?

ML Engineering (MLOps) vs AI Engineering

Technical Comparison (Chapter 01)

Feature	Traditional MLOps	AI Engineering
Primary Objective	Building, training, and deploying task-specific models from scratch	Building applications on top of readily available foundation models
Data Focus	Massive amounts of labeled data, heavy feature engineering and tabular data processing	Focus on context construction, data curation, and data synthesis
Core Techniques	Model architecture design, full-parameter training, and supervised learning	Prompt engineering, Retrieval-Augmented Generation (RAG), and PEFT (Parameter-Efficient Finetuning)
Optimization Goal	Minimizing training loss and maximizing specific task metrics (e.g., F1, Accuracy)	Optimizing for latency, cost, and reliability of the end-to-end application
Infrastructure	Large-scale training clusters and complex model-serving pipelines	Model-as-a-Service APIs or efficient local hosting of pre-trained weights
Iteration Cycle	Months-long cycles: data collection, labeling, training, eval	Rapid cycles: weekend projects using few-shot examples and prompt iteration
Evaluation	Static benchmarks and well-defined metrics like Cross-Entropy	Complex and "hardest" challenge, involves AI-as-a-judge and open-ended qualitative assessment
System Logic	Deterministic pipelines based on specific model outputs	Probabilistic system design; managing inconsistency and hallucinations

Committee of Supply 2026

Building Singapore's digital capability in a digital age.
IMDA's priorities focus on thriving together in the digital
age and building our digital community.

Learn more >

01

Scroll for more

Be digitally empowered for innovation and growth

At IMDA, we empower businesses to leverage opportunities brought forth by digitalisation, through expert digital transformation strategy consulting, digital innovation and solutions that will aid them in integrating tech, while strengthening their digital foundations.

Starter Kit for Testing LLM-Based Applications for Safety and Reliability

Building a Trusted, Secure and Reliable AI Ecosystem

Version 1.0
January 2026

Table of Contents

Executive Summary	06
Part 1: Introduction	08
1.1 Objectives and Scope	09
1.1.1 What Does the Starter Kit Do	09
1.1.2 Who is the Starter Kit For	09
1.1.3 When to Use the Starter Kit	10
1.1.4 How to Read the Starter Kit Alongside Other Frameworks	10
1.2 Baseline Safety and Reliability	11
1.2.1 What is Baseline Safety and Reliability	11
1.2.2 What are Core Benchmarks	11
1.3 Structured Approach to LLM App Testing	13
1.3.1 How to Ensure that the Tests are Rigorous and Trusted	13
1.3.2 What is Output Testing	13
1.3.3 What is Component Testing	13
1.3.4 Ensuring Robustness and Consistency of Results	15
Part 2: Foundational Concepts	16
2.1 Benchmarks	17
2.1.1 What is a Benchmark	17
2.1.2 What Makes a "Good" Test Dataset	19
2.1.3 What Makes a "Good" Metric	21
2.1.4 What Makes a "Good" Evaluator	22
2.2 Red Teaming	24
2.2.1 What is Red Teaming	24
2.2.2 What is "Good" Red Teaming	25
2.2.3 What is Automated Red Teaming	27

Part 3: Structured Testing Approach	28
STEP 1: IDENTIFY RELEVANT RISKS AND SET THRESHOLDS	30
3.1.1 Understanding Key Risks	30
3.1.2 Identifying Relevant Risks	31
3.1.3 Calibrating the Extent of Testing	34
3.1.4 Setting the Thresholds	34
STEP 2: TEST FOR THE RELEVANT RISKS	37
Hallucination and Inaccuracy	37
3.2.1 Output Testing: Domain-specific Knowledge	38
3.2.2 Output Testing: Out-of-Domain Topics	45
3.2.3 Component Testing: Testing Retrieval-Augmented Generation	47
Bias in Decision Making	52
3.3.1 Output Testing: Parity Testing	54
3.3.2 Output Testing: Perturbation Testing	59
3.3.3 Component Testing: Model and Other App Components	60
Undesirable Content	62
3.4.1 Output Testing: Types of Undesirable Content and Ease of Elicitation	65
3.4.2 Output Testing: Helpfulness of Responses	68
3.4.3 Component Testing: Input and Output Filters	69
Data Leakage	73
3.5.1 Output Testing: Types of Sensitive Data and Ease of Elicitation	76
3.5.2 Output Testing: Helpfulness of Responses	77
3.5.3 Component Testing: System Prompt	78
Vulnerability to Adversarial Prompts	82
3.6.1 Output Testing: Direct Prompt Attacks	84
3.6.2 Output Testing: Indirect Prompt Injections	90
3.6.3 Component Testing: Input Filters	91
STEP 3: ANALYSE RESULTS	94
3.7.1 Determining Whether Baseline Safety and Reliability Have Been Achieved	94
3.7.2 Analyse and Interpret Results	94
3.7.3 Continuous Monitoring and Re-testing	96

Agosto - 2026						
D	S	T	Q	Q	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

Setembro - 2026						
D	S	T	Q	Q	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

Outubro - 2026						
D	S	T	Q	Q	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

Novembro - 2026						
D	S	T	Q	Q	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

Dezembro - 2026						
D	S	T	Q	Q	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

Unit 01

Week 01

10/08 – Re-Enroll/Planing

12/08 – Course Outline

Week 02

17/08 – Visualizing Gradient Descendent

19/08 – Rethinking the training loop

Week 03

24/08 – A simple classification problem

26/08 – From a shallow to a deep-ish classification model

Week 04

31/08 – LLM Introduction (History, Embedding)

02/09 – Architecture Overview

Week 05

07/09 – No class

09/09 – Ollama

Week 06

14/09 – RAG

16/09 – Advance RAG

Weeks 07, 08, 09, 10, 11 (Hands on Project)

21/09 – Project

23/09 – Project

28/09 – Project

30/09 – Project

05/10 – Project

07/10 – Project

12/10 – No class

14/10 – Review/Consulting

19/10 – Project Presentation (paper, slides, repo)

21/10 – Project Presentation (paper, slides, repo)

End of Unit 01 (21/10)

Agosto - 2026						
D	S	T	Q	Q	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

Setembro - 2026						
D	S	T	Q	Q	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

Outubro - 2026						
D	S	T	Q	Q	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

Novembro - 2026						
D	S	T	Q	Q	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

Dezembro - 2026						
D	S	T	Q	Q	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

Unit 02 and 03

Week 12

26/10 – Agents I (Introduction, architectures)

28/10 – No class

Week 13

02/11 – No class

04/11 – Agents II – MCP

Week 14

09/11 – Agents III – Frameworks

11/11 – Claude Ecosystem I

Week 15

16/11 – Claude Ecosystem II

18/11 – Claude Ecosystem III

Week 16

23/11 – Claude Ecosystem IV

25/11 – Claude Ecosystem V

Weeks 16, 17 and 18 (Hands on Project)

30/11 – Project Description

02/12 – Project

07/12 – Project

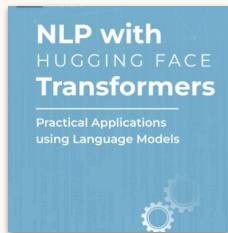
09/12 – Project

14/12 – Project Presentation

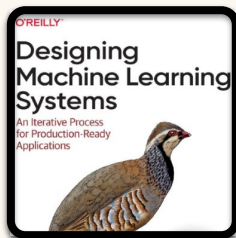
16/12 – Project Presentation

End of Unit 02 and 03 (16/12)

References



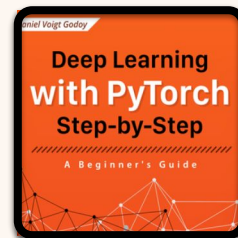
May, 2025



May, 2022



Jan, 2025



Feb, 2022

COHORT-BASED COURSE

AI Coding for Real Engineers

A two-week cohort for developers who want to use AI coding tools on real production-style work: context gathering, planning, steering, feedback loops, AFK agents, and human-in-the-loop review.



Matt Pocock

A well-regarded expert developer known for his ability to demystify complex concepts.

 **UDACITY**
Part of Accenture

Discover ▾ Catalog Individuals Business ▾ Government AI Master's ▾

AI Programs > School Of Artificial Intelligence > AI Engineering with Claude

AI Engineering with Claude

The AI Engineering with Claude Nanodegree program prepares developers to build and ship production-grade AI agent systems with Claude, the Claude Agent SDK, the Model Context Protocol (MCP), and Claude Code. Across four project-based courses, you'll go beyond demos to engineer the "harness" that runs an agent in production: controlling the agentic loop, governing MCP tools, evaluating systems so silent failures surface before customers do, and enforcing rules in code with bounded autonomy and human-in-the-loop guardrails. Portfolio projects are anchored in real industry scenarios from insurance, retail, financial services, and manufacturing, including an MCP-powered "Agentic Analyst" and an enterprise multi-agent code-review orchestrator.

 **Nanodegree Program** = Intermediate ⌚ 38 hours 📅 Updated: Jul 29, 2026

Clone me!!!!

<https://github.com/ivanovitchm/aiengineering>